CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 13: Section and Mid-Point Formula

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Centroid of a triangle

1. Triangle ABC has vertices $A(0, 6)$, $B(4, -4)$ and $C(-4, -2)$. D is the point where AC crosses the x-axis. (a) Find the coordinates of D. (b) Find the centroid of $\triangle ABC$. (c) D divides AC in the ratio $k : 1$; find k. (d) Find the equation of the line BD. ICSE 2024, 4 marks

SOLUTION

(c) Let D divide AC in the ratio $k : 1$. Its y-coordinate is 0: $\dfrac{-2k + 6}{k + 1} = 0$, so $k = 3$.

(a) Then $x = \dfrac{3(-4) + 0}{3 + 1} = -3$, so $D = (-3, 0)$.

(b) Centroid $= \left(\dfrac{0 + 4 - 4}{3}, \dfrac{6 - 4 - 2}{3}\right) = (0, 0)$

(d) Slope of $BD = \dfrac{0 - (-4)}{-3 - 4} = -\dfrac{4}{7}$. Through D: $y = -\dfrac{4}{7}(x + 3)$, so $7y = -4x - 12$.

Answer: (a) $D(-3, 0)$ **(b)** $(0, 0)$ **(c)** $k = 3$ **(d)** $4x + 7y + 12 = 0$

2. The vertices of $\triangle ABC$ are $(-4, -2)$, $(6, 2)$ and $(4, 6)$. Find its centroid. (a) $(2, 2)$ (b) $(2, 3)$ (c) $(3, 3)$ (d) $(0, -1)$ ICSE 2023, 1 mark

SOLUTION

Centroid $= \left(\dfrac{-4 + 6 + 4}{3}, \dfrac{-2 + 2 + 6}{3}\right) = (2, 2)$

Answer: (a)

2 The section formula

3. $A(4, 0)$ lies on the x-axis and $B(0, 4)$ lies on the y-axis. P is the point on AB with $AP : PB = 3 : 1$. (a) Use the section formula to find P. (b) Find the equation of the line through P perpendicular to AB. ICSE 2023, 4 marks

SOLUTION

(a) $P = \left(\dfrac{3 \times 0 + 1 \times 4}{3 + 1}, \dfrac{3 \times 4 + 1 \times 0}{3 + 1}\right) = (1, 3)$

(b) Slope of $AB = \dfrac{4 - 0}{0 - 4} = -1$, so the perpendicular line has slope 1.

$y - 3 = 1(x - 1)$

Answer: (a) $P(1, 3)$ **(b)** $y = x + 2$

4. M lies on the x-axis and N lies on the y-axis. The point $P(3, 2)$ divides MN in the ratio $2 : 3$. Find (a) the coordinates of M and N (b) the slope of MN. ICSE 2019, 3 marks

SOLUTION

(a) Let $M = (a, 0)$ and $N = (0, b)$. Then $P = \left(\frac{2 \times 0 + 3a}{5}, \frac{2b + 3 \times 0}{5}\right)$.

$\frac{3a}{5} = 3$ gives $a = 5$, and $\frac{2b}{5} = 2$ gives $b = 5$.

(b) Slope of $MN = \frac{5 - 0}{0 - 5} = -1$

Answer: (a) $M(5, 0)$, $N(0, 5)$ **(b) slope** $= -1$

5. $P(1, -2)$ lies on the segment joining $A(3, -6)$ and $B(x, y)$, with $AP : PB = 2 : 3$. Find the coordinates of B. ICSE 2017, 4 marks

SOLUTION

$1 = \frac{2x + 3(3)}{5}$, so $2x = -4$ and $x = -2$.

$-2 = \frac{2y + 3(-6)}{5}$, so $2y = 8$ and $y = 4$.

Answer: $B(-2, 4)$

3 Finding the ratio

6. In what ratio does the y-axis divide the segment joining $P(5, 3)$ and $Q(-5, 3)$? Find the point of intersection. ICSE 2020, 4 marks

SOLUTION

Let the ratio be $k : 1$. On the y-axis the x-coordinate is 0: $\frac{-5k + 5}{k + 1} = 0$, so $k = 1$.

The point is the mid-point: $\left(0, \frac{3 + 3}{2}\right) = (0, 3)$.

Answer: ratio $1 : 1$**; the point is** $(0, 3)$

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.